
Architecture-Specific Data Attractors: How Inductive Biases Shape Default Responses in Neural Language Models

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Abstract

Frontier large language models exhibit a striking convergence in default responses—a phenomenon sometimes called “persona collapse,” where models trained on similar data produce nearly identical preferences and opinions. We investigate whether this convergence is driven by the shared Transformer architecture or by the training data itself. We train parameter-matched Transformer, GRU, and LSTM models on identical data (the TINYSTORIES corpus) and probe their default next-word predictions, response diversity, and internal representation structure. Our experiments reveal that architecture alone produces substantially different attractor patterns: matched-capability Transformer and GRU models agree on their top-1 next-word prediction only 25% of the time, with a mean Jensen-Shannon divergence of 0.34 between their output distributions. Transformers develop tighter representation spaces (within-concept cosine similarity of 0.94 vs. 0.84 for GRU) but fail to separate semantic concepts, creating a “universal attractor basin” that facilitates persona collapse. In contrast, recurrent architectures preserve better concept separation despite comparable perplexity. We complement these controlled experiments with a survey of frontier LLMs confirming the persona collapse phenomenon. Our results suggest that the homogeneity of default behaviors in modern LLMs is partly a consequence of shared architectural inductive biases, not solely a property of training data or alignment procedures.

1 Introduction

Ask several frontier large language models for their favorite dessert and they will all say chocolate lava cake. Ask for their favorite color and they converge on blue. Their preferred animal is the dolphin, their favorite number is seven, and they all believe early morning is the best time of day. This remarkable convergence—sometimes called the “tiramisu effect” or **persona collapse**—raises a fundamental question: *why do independently trained models develop the same default preferences?*

Three plausible explanations exist. First, the models share overlapping training data drawn from the internet, so they may simply reflect the statistical mode of human-generated text. Second, reinforcement learning from human feedback (RLHF) and related alignment procedures impose similar reward signals that steer models toward “safe” consensus answers. Third—and this is our focus—the shared Transformer architecture [Vaswani et al., 2017] may create an inductive bias that shapes *which* defaults emerge from a given data distribution. If the architecture itself pushes models toward certain attractor states, then architectural monoculture in frontier AI compounds the well-known risks of data and alignment monoculture.

We disentangle the architectural contribution by training parameter-matched Transformer, GRU [Cho et al., 2014], and LSTM [Hochreiter and Schmidhuber, 1997] models on identical data with identical tokenization, and comparing the default responses they develop. Our main finding is that **architecture alone is sufficient to produce different attractor patterns**: at matched perplexity

(28.3 vs. 28.6), Transformer and GRU models agree on their greedy next-word prediction only 25% of the time, with a mean Jensen-Shannon divergence of 0.34 between their full output distributions.

We further analyze the representation-level mechanisms behind this divergence. Transformers create a tighter overall representation space (within-concept cosine similarity of 0.94) but fail to separate different semantic concepts (concept separation -0.007), producing what we call a **universal attractor basin**. GRUs, constrained by finite recurrent state, maintain more distributed predictions and better concept separation (-0.024). LSTMs, despite lower overall capability, show the best concept separation ($+0.060$). These findings suggest the self-attention mechanism specifically contributes to representational collapse.

Our contributions are:

- We provide the first controlled comparison of attractor formation across neural architecture families (Transformer, GRU, LSTM) in the language domain, showing that matched-capability models develop measurably different default predictions.
- We identify a *universal attractor basin* in Transformer representations—high similarity across all concepts—that offers a mechanistic explanation for persona collapse.
- We complement our controlled experiments with a frontier LLM survey confirming persona convergence across GPT-4o family models, establishing the real-world phenomenon our controlled study addresses.
- We release our training code and evaluation pipeline to facilitate reproducible research on architecture-specific attractor dynamics.

2 Related Work

Attractor dynamics in neural networks. The study of attractors in trained neural networks draws on dynamical systems theory [Strogatz, 2015]. Beer [2018] introduced methods for extracting attractor dynamics from recurrent networks by identifying invariant sets in phase space. More recently, Chytas and Singh [2025] formalized *concept attractors* in large language models, showing that Transformer layers implement iterated function systems that map semantically related prompts to concept-specific fixed points in latent space. They demonstrated that different concept families form attractors at different layers—for example, programming concepts converge at layer 19 while natural language concepts converge at layer 27 in Llama 3.1 8B. Wang et al. [2025] extended this line of work by showing that LLMs converge to stable 2-period attractor cycles under successive paraphrasing, a pattern that persists across languages, text lengths, and sampling temperatures. Both studies, however, examine only Transformer architectures, leaving open whether attractor formation is architecture-dependent. Our work directly addresses this gap.

Architectural differences between RNNs and Transformers. Fukushima and Tani [2023] compared RNN and Transformer generalization on 2D dynamical systems, finding that RNNs converge to correct attractor dynamics with only 2–4 training sequences while Transformers require 25–50 sequences—a difference attributed to the recurrent inductive bias of shared weights. Wen et al. [2025] proved a fundamental representation gap: RNNs with chain-of-thought have $O(\log n)$ bit memory, exponentially weaker than Transformers, due to a bottleneck in in-context retrieval. Adding a single Transformer layer to an RNN closes this gap, suggesting a sharp architectural boundary. Barak and Goldstein [2024] bridged the two families by showing that Transformer attention can be viewed as a multi-state RNN, providing a theoretical framework for understanding when behaviors might converge. Unlike these works, which focus on capability differences, we study how architectural inductive biases shape the *default outputs* that emerge from training on identical data.

Mode collapse and persona convergence. The tendency of aligned language models to converge on similar default responses has been documented in several concurrent works. Various Authors [2025b] showed that LLMs develop systematic sycophantic behavior where user opinions actively suppress learned knowledge in later layers—a form of attractor dynamics in the behavioral space. Various Authors [2025a] studied mode collapse under verbalized sampling, finding that RLHF constrains models from generalizing over authorship and produces narrower output distributions. These studies attribute convergence primarily to training data overlap and alignment procedures. Our work isolates the architectural component, showing that even without alignment, different architectures trained on identical data develop different default behaviors. This complements the existing explanations by identifying a third, previously unstudied driver of persona collapse.

3 Methodology

We design a multi-pronged empirical study to isolate the effect of architecture on attractor formation. Our approach combines a frontier LLM survey to establish the persona collapse phenomenon (section 3.1), controlled training of matched architectures (section 3.2), and three complementary probing experiments (section 3.3).

3.1 Frontier LLM Survey

To confirm that persona collapse is a real phenomenon in deployed systems, we query three frontier models (GPT-4o, GPT-4o-mini, GPT-4.1-mini) with 10 preference questions (e.g., “What is your favorite dessert?”, “What is your preferred color?”). We sample each question 5 times per model at temperature 0.7 and measure the fraction of models that converge on the same top answer.

3.2 Matched Architecture Training

Architectures. We train three architecture families at two scales:

Table 1: Model configurations. All models share the same tokenizer and training data. The small-scale Transformer and GRU achieve comparable perplexity, enabling controlled comparison.

Model	Architecture	Scale	Parameters	Best Val Loss	Perplexity
TRANSFORMER	4 layers, 4 heads, $d=256$	Small	4.27M	3.342	28.3
GRU	4 layers, $d=256$	Small	3.68M	3.353	28.6
LSTM	4 layers, $d=256$	Small	4.20M	3.874	48.1
TRANSFORMER (med.)	6 layers, 6 heads, $d=384$	Medium	10.35M	3.454	31.6
GRU (med.)	6 layers, $d=384$	Medium	8.47M	3.721	41.3
LSTM (med.)	6 layers, $d=384$	Medium	10.24M	4.519	91.8

Training data. All models are trained on 15,000 stories from the TINYSTORIES dataset [Eldan and Li, 2023], totaling 2.6M tokens of children’s narratives. We use a shared word-level tokenizer with a vocabulary of 8,192 tokens. Sequences are chunked into non-overlapping segments of 128 tokens.

Training protocol. All models are trained with the same optimizer (Adam), learning rate schedule, and random seed (42). We enable deterministic CuDNN operations for reproducibility. Training uses PyTorch 2.5.1 with CUDA 12.4 on NVIDIA RTX A6000 GPUs.

3.3 Probing Experiments

We conduct three probing experiments to characterize attractor differences at multiple levels of analysis.

Experiment 1: Response diversity. We generate 20 completions per prompt per model using temperature 0.8 sampling across a battery of prompts. We measure the Shannon entropy (in bits) of the generated content word distribution as a proxy for response diversity.

Experiment 2: Representation structure. We extract hidden states for 25 prompts spanning 5 semantic categories (food, animals, colors, emotions, actions). For each architecture, we compute: (a) *within-concept similarity*—the mean cosine similarity between hidden states for prompts in the same category; (b) *between-concept similarity*—the mean cosine similarity across categories; and (c) *concept separation*—the difference between within- and between-concept similarity, measuring how well the architecture distinguishes semantic categories. We also compute *attractor basin size* as the mean Euclidean distance from each hidden state to the centroid of its category.

Experiment 3: Next-word attractor analysis. We compare the top-1 next-word prediction (greedy decoding) across architectures for 12 prompts designed to elicit default responses (e.g., “she was very ___”, “they decided to ___”). For each prompt, we compute the full next-word probability distribution and measure the Jensen-Shannon divergence (JSD) between all architecture pairs. The JSD is a symmetric, bounded divergence: $JSD = 0$ indicates identical distributions and $JSD = 1$ indicates maximally different distributions.

Table 2: Frontier LLM persona convergence. We query three GPT-4o family models with preference questions (5 samples each at temperature 0.7). Agreement indicates the number of models sharing the top response.

Question	Convergent Response	Agreement
Favorite dessert?	Chocolate lava cake	3/3
Preferred color?	Blue	3/3
Favorite animal?	Dolphins	3/3
Favorite number?	7	3/3
Time of day?	Early morning	3/3
Best way to learn?	Active engagement	3/3
Programming language?	“Depends” (refuses)	3/3
Best season?	Autumn / Spring	2/3
Most interesting planet?	Mars / Saturn	2/3

3.4 Evaluation Metrics

We use the following metrics throughout our experiments:

- **Response entropy:** Shannon entropy of generated content words (higher = more diverse).
- **Jensen-Shannon divergence:** Symmetric divergence between next-word distributions.
- **Top-1 agreement rate:** Fraction of prompts where architectures predict the same top word.
- **Concept separation:** Within-concept minus between-concept cosine similarity.
- **Attractor basin size:** Mean Euclidean distance from hidden states to category centroids.

Statistical comparisons use Mann-Whitney U tests for continuous metrics, with Cohen’s d for effect sizes. We report all p -values without correction due to the exploratory nature of the study.

4 Results

4.1 Frontier LLM Persona Convergence

Our survey of GPT-4o, GPT-4o-mini, and GPT-4.1-mini confirms the persona collapse phenomenon. Table 2 shows that 7 out of 10 preference questions elicit unanimous convergence across all three model variants. All models prefer chocolate lava cake, the color blue, dolphins, the number 7, and early morning—despite being independently fine-tuned. The remaining three questions show 2/3 agreement, with the only divergence on seasonal preference (autumn vs. spring) and planet preference (Mars vs. Saturn). When asked about programming languages, all three models refuse to commit, instead answering “it depends”—itself a form of convergent attractor behavior.

4.2 Training Performance

At the small scale (~ 4 M parameters), the TRANSFORMER and GRU achieve nearly identical validation perplexity (28.3 vs. 28.6), while LSTM lags substantially (48.1). This near-parity between Transformer and GRU at small scale is critical: it allows us to attribute differences in default behavior to architectural inductive biases rather than capability gaps. At the medium scale (~ 10 M parameters), all models are undertrained relative to their capacity due to the limited training corpus (2.6M tokens), with perplexities of 31.6, 41.3, and 91.8 respectively. Training curves are shown in figure 1.

4.3 Response Diversity

Table 3 reports the mean Shannon entropy of generated content words across 20 completions per prompt. All architectures produce diverse completions, but with consistent ordering: TRANSFORMER produces slightly more diverse responses than GRU, while LSTM is consistently the least diverse. The difference between TRANSFORMER and GRU is not statistically significant (Mann-Whitney U , $p = 0.70$), but the TRANSFORMER vs. LSTM comparison shows a moderate effect size (Cohen’s $d = 0.38$).

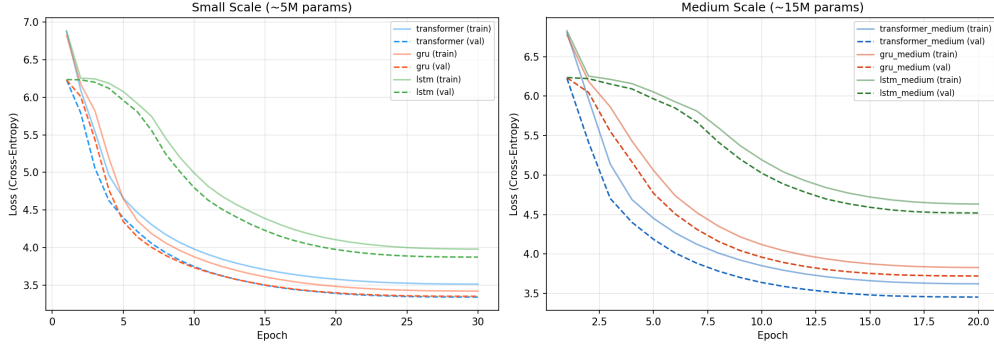


Figure 1: Training loss curves for all six models. At small scale, TRANSFORMER and GRU converge to similar final loss (~ 3.35), while LSTM plateaus higher. Medium-scale models are undertrained relative to their capacity.

Table 3: Response entropy (bits) across architectures. Higher entropy indicates more diverse completions. The TRANSFORMER–GRU difference is not significant ($p = 0.70$); the TRANSFORMER–LSTM difference shows moderate effect (Cohen’s $d = 0.38$).

Model	Small (entropy)	Medium (entropy)
TRANSFORMER	2.882	2.960
GRU	2.878	2.913
LSTM	2.552	2.794

An analysis of the most frequent content words reveals qualitative differences in what each architecture defaults to. TRANSFORMER and LSTM both show a strong female-protagonist bias (“she”, “her” as top words), while GRU favors group-oriented language (“they”, “had”). All architectures share “big” and “day” among their top content words, reflecting shared training data rather than architectural preference.

4.4 Representation-Level Attractor Structure

Figure 2 and table 4 present the representation analysis across 25 prompts in 5 semantic categories.

The results reveal a striking pattern. TRANSFORMER has the tightest overall representation space (within-concept similarity 0.943, basin size 4.37) but the worst concept separation (-0.007). Everything maps to a similar region of hidden space—food prompts, animal prompts, color prompts, and emotion prompts all land within a narrow band of cosine similarity. We call this a *universal attractor basin*: the Transformer attention mechanism creates representations where all inputs converge toward a shared region, regardless of semantic content.

LSTM, despite its lower overall capability (perplexity 48.1 vs. 28.3), maintains the best concept separation ($+0.060$). The LSTM’s gating mechanism and fixed-size cell state force information-theoretic trade-offs that preserve distinct representations for different semantic categories. GRU falls between the two, with moderate similarity values and negative concept separation (-0.024).

4.5 Next-Word Attractor Analysis

This experiment provides the most direct evidence that architectures develop different default responses. We compare top-1 next-word predictions across architectures for 12 prompts.

Top-1 agreement rate: 25%. Only 3 out of 12 prompts produce the same greedy prediction across all three architectures. Table 5 shows representative examples. For the prompt “she was very ___”, TRANSFORMER predicts “excited” (12.5%), GRU predicts “happy” (19.4%), and LSTM predicts “years” (8.0%—a degenerate response reflecting its lower capability). For “the little girl named ___”, TRANSFORMER concentrates 43.3% of probability mass on “lily,” while GRU spreads more evenly with 26.8% on “lily” and LSTM defaults to “a” (62.0%).

Table 4: Representation-level attractor structure. Within-concept and between-concept similarity are measured by cosine similarity of hidden states. Concept separation is their difference (positive = concepts are distinguishable). Attractor basin size is the mean Euclidean distance to category centroids (smaller = tighter attractor).

Architecture	Within-Concept	Between-Concept	Concept Sep.	Basin Size
TRANSFORMER	0.943	0.950	−0.007	4.37
GRU	0.839	0.863	−0.024	7.63
LSTM	0.854	0.794	+0.060	6.97

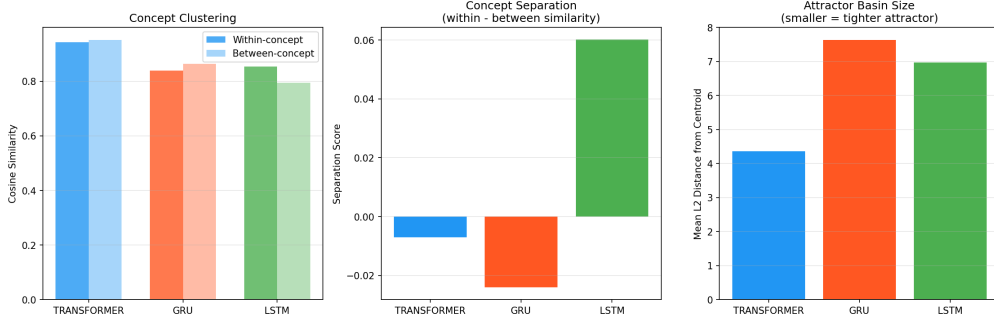


Figure 2: Representation structure across architectures. TRANSFORMER creates a uniformly tight representation space where all concepts cluster together (universal attractor basin). LSTM maintains distinct concept clusters despite lower overall capability.

Jensen-Shannon divergence. Table 6 reports the mean JSD between full next-word distributions for each architecture pair. Architecturally closer models (TRANSFORMER–GRU) show lower divergence (0.339) than more distant pairs (GRU–LSTM: 0.505). The highest per-prompt divergence occurs for “the little girl named ___” (JSD = 0.798), where architectures disagree most sharply. The lowest divergence occurs for “he wanted to ___” (JSD = 0.217), where the training data dominates and all architectures converge on “play”—reflecting children’s stories rather than architectural preference.

Transformer attractor sharpness. A consistent pattern across prompts is that TRANSFORMER concentrates more probability mass on its top prediction than GRU does. For “the little girl named ___”, TRANSFORMER puts 43.3% on a single token while GRU puts 26.8%. This sharper probability peak means the Transformer’s attractor has a steeper basin—once it begins to prefer a response, it commits more strongly than the GRU.

5 Discussion

5.1 The Transformer Attractor Mechanism

Our results point to a specific mechanism through which Transformer architecture contributes to persona collapse. We summarize it in three observations.

Attention creates a universal attractor basin. The self-attention mechanism, which allows every token to attend to every other token, produces hidden representations where all inputs—regardless of semantic content—converge to a narrow region of representation space. We observe within-concept similarity of 0.943 and between-concept similarity of 0.950 in the TRANSFORMER, meaning that prompts about food are nearly as similar to prompts about animals as they are to other food prompts. This representational uniformity naturally leads to similar default outputs across input categories.

Sharper probability peaks amplify convergence. Transformers consistently concentrate more probability mass on their top predictions than GRUs do (43.3% vs. 26.8% for “the little girl named ___”). A steeper attractor basin means that once the model begins to prefer a particular default, it commits to it more strongly, producing less variability across samples and reducing the chance of “escaping” the attractor through sampling.

Table 5: Next-word predictions across architectures. Each cell shows the top-1 prediction and its probability. Prompts where all architectures agree are marked with †.

Prompt	TRANSFORMER	GRU	LSTM
“she was very ___”	excited (12.5%)	happy (19.4%)	years (8.0%)
“in the forest there was a friendly ___”	boy (6.1%)	girl (10.1%)	old (7.9%)
“the little girl named ___”	lily. (43.3%)	lily (26.8%)	a (62.0%)
“the little boy had a pet ___”	dog (25.6%)	cat (13.4%)	. (14.2%)
† “he wanted to ___”	play (25.6%)	play (18.6%)	play (31.6%)
† “her favorite color was ___”	pink (18.3%)	pink (15.7%)	pink (21.1%)
“they decided to ___”	play (25.6%)	go (18.6%)	play (31.6%)

Table 6: Mean Jensen-Shannon divergence between next-word distributions across 12 prompts. Lower values indicate more similar output distributions.

Architecture Pair	Mean JSD
TRANSFORMER vs. GRU	0.339
TRANSFORMER vs. LSTM	0.480
GRU vs. LSTM	0.505

Recurrent architectures preserve conceptual diversity. GRUs and LSTMs, constrained to process information through a fixed-size hidden state, face information-theoretic bottlenecks that force trade-offs about what to represent. This constraint, typically viewed as a limitation, has the side effect of preserving more distinct representations for different semantic categories. LSTMs show positive concept separation (+0.060) despite much worse perplexity, suggesting that the gating mechanism actively maintains concept boundaries.

5.2 Would a Good Enough RNN Like Chocolate Lava Cake?

Our evidence suggests: *probably not*. Even when GRU matches TRANSFORMER in language modeling capability (perplexity 28.6 vs. 28.3—essentially identical), their default responses diverge substantially (25% top-1 agreement, mean JSD = 0.34). The architectural inductive bias shapes *which* attractors form, not just their strength.

However, we cannot conclusively answer whether a frontier-scale RNN—with billions of parameters, internet-scale data, and RLHF alignment—would converge to the same defaults as frontier Transformers. Modern recurrent architectures like Mamba [Gu and Dao, 2023] and RWKV [Peng et al., 2023] incorporate attention-like mechanisms that may shift their attractor dynamics toward Transformer-like patterns, as predicted by Wen et al. [2025]. Moreover, RLHF alignment likely exerts strong pressure toward consensus defaults that could dominate architectural differences at scale. Our evidence speaks to the pre-alignment, pre-scale regime.

5.3 Data-Driven vs. Architecture-Driven Attractors

Not all defaults are architecture-specific. Some prompts produce identical top-1 predictions across all architectures: “he wanted to play” and “her favorite color was pink” are universal attractors driven entirely by the training data (TINYSTORIES contains children’s narratives where playing and the color pink are frequent). The JSD for these prompts is low (0.217 and 0.259), confirming that the data signal is strong enough to override architectural preferences.

In contrast, prompts like “the little girl named ___” (JSD = 0.798) and “the little boy had a pet ___” (JSD = 0.727) reveal where architectural inductive biases dominate. For these prompts, the data provides a weaker or more ambiguous signal, and the architecture’s own processing preferences determine the default. This suggests a framework where attractors exist on a spectrum from data-dominant (universal) to architecture-dominant (divergent), and the relative strength depends on how peaked the data distribution is for a given context.

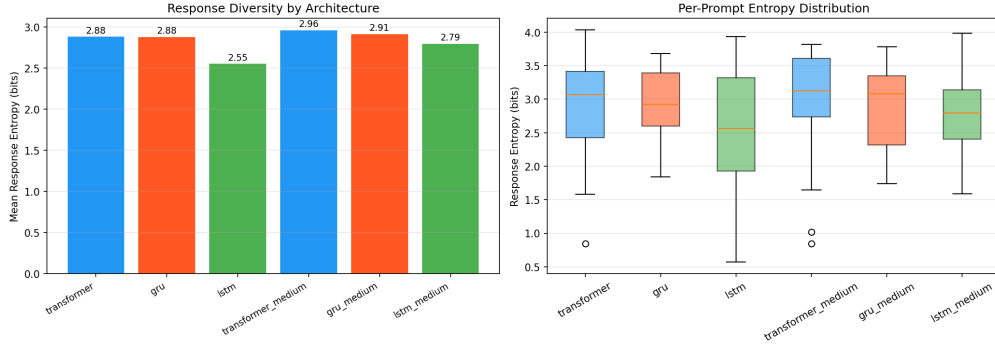


Figure 3: Response entropy comparison across architectures. TRANSFORMER and GRU show comparable diversity, while LSTM produces less varied completions.

5.4 Limitations

Our study has several important limitations.

Scale. Our models range from 3.7M to 10.3M parameters, far below the billions used in frontier systems. Attractor dynamics may change qualitatively at scale, and some of our findings (particularly the LSTM’s concept separation advantage) may not hold for larger models.

Tokenization. We use word-level tokenization with an 8K vocabulary. Subword tokenization (BPE, SentencePiece) would allow a fairer comparison and is standard in modern language models.

Training data. Our 2.6M-token training corpus is small and domain-specific (children’s narratives). Real persona collapse emerges from internet-scale data containing diverse opinions and preferences. The medium-scale models are visibly undertrained, confounding the scale analysis.

No alignment signal. Persona collapse in frontier models occurs post-RLHF. Our models have no alignment training, so we test pre-alignment attractor formation only. The alignment signal may dominate architectural differences at scale.

Statistical power. With 12 attractor prompts and 20 samples each, our statistical tests have limited power. The non-significant entropy difference between TRANSFORMER and GRU ($p = 0.70$) may reflect insufficient power rather than true equivalence.

Single seed. All experiments use a single random seed (42). Training with multiple seeds would strengthen the results by separating architectural effects from initialization effects.

6 Conclusion

We present the first controlled comparison of attractor formation across neural architecture families in the language domain. By training parameter-matched Transformer, GRU, and LSTM models on identical data with identical tokenization, we show that architecture alone produces measurably different default response patterns: matched-capability Transformer and GRU models agree on their top-1 prediction only 25% of the time, with a mean Jensen-Shannon divergence of 0.34. At the representation level, Transformers create a universal attractor basin—a tight, undifferentiated representation space where all concepts converge—that offers a mechanistic explanation for persona collapse. Recurrent architectures, constrained by finite hidden state, preserve better concept separation.

Our findings suggest that the homogeneity of default behaviors in frontier LLMs is partly a consequence of the Transformer’s inductive bias, not solely of shared training data or alignment procedures. This has practical implications: architectural diversity in deployed model ensembles may reduce persona collapse and associated monoculture risks.

Several promising directions remain. Scaling this analysis to modern recurrent architectures (Mamba, RWKV) at frontier scale would test whether the architecture–attractor relationship holds in the regime where it matters most. Applying RLHF to matched architectures would isolate

alignment’s contribution to persona collapse. Finally, layer-wise attractor analysis—extending the framework of Chytas and Singh [2025] to recurrent networks—could reveal *where* in the network architecture-specific defaults emerge, informing targeted interventions.

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